Name: J. Renuka Devi

Reg.no: 192573007

Course code: DSA0602

Course name: Data Handling and Visualization

Class Activity 1 UNIT-2 20.07.2026

R Programming

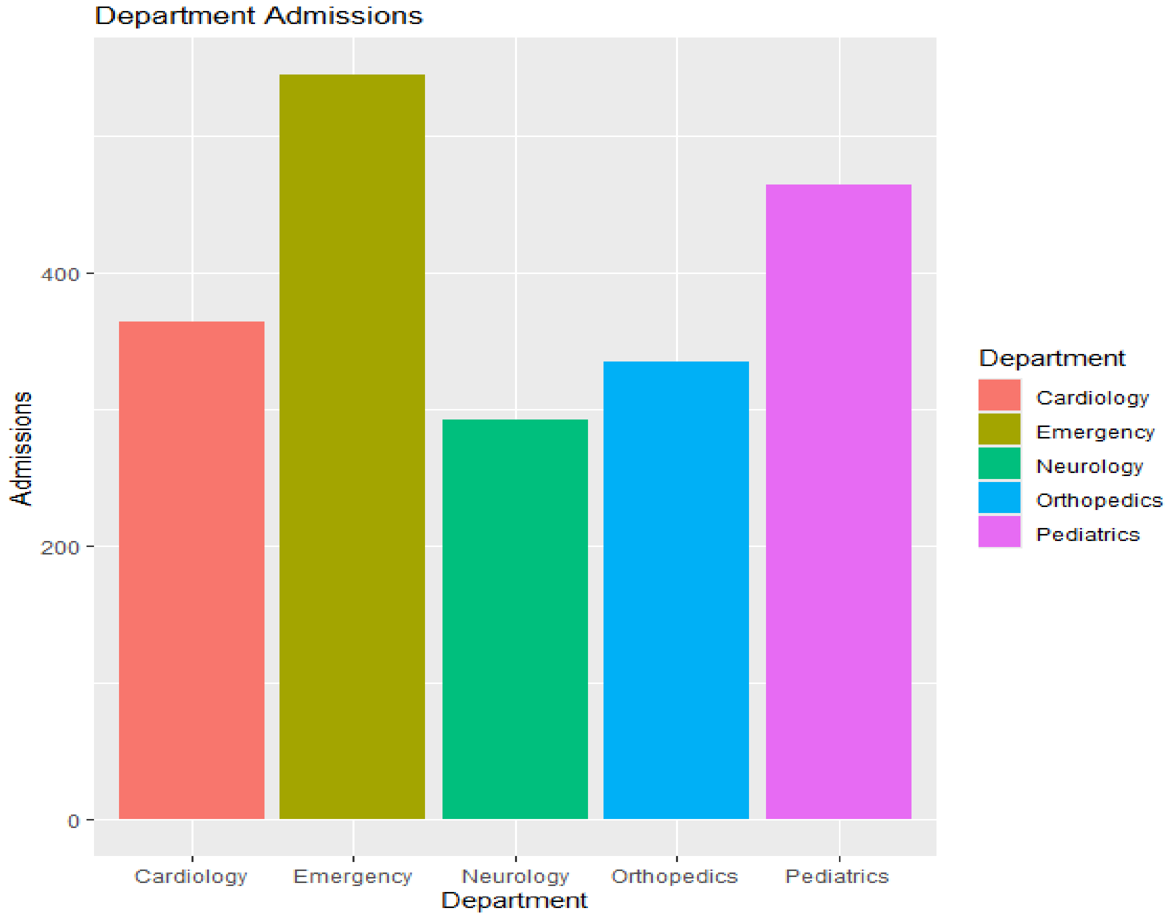
SCENARIO 1. HOSPITAL PATIENT ADMISSION ANALYSIS

DATASET:

```
library(ggplot2)
library(treemap)
hospital <- data.frame(
  Department = c(
    "Cardiology","Cardiology","Cardiology","Cardiology","Cardiology",
    "Neurology","Neurology","Neurology","Neurology","Neurology",
    "Pediatrics","Pediatrics","Pediatrics","Pediatrics","Pediatrics",
    "Orthopedics","Orthopedics","Orthopedics","Orthopedics","Orthopedics",
    "Emergency","Emergency","Emergency","Emergency","Emergency"
  ),
  Gender = c(
    "Male","Female","Male","Female","Male",
    "Male","Female","Male","Female","Male",
    "Male","Female","Male","Female","Male",
    "Male","Female","Male","Female","Male",
    "Male","Female","Male","Female","Male"
  ),
  Admissions = c(
    120,125,118,130,122,
    95,100,98,102,97,
    150,160,155,148,158,
    110,115,108,120,112,
    180,185,175,190,182
  )
)
```

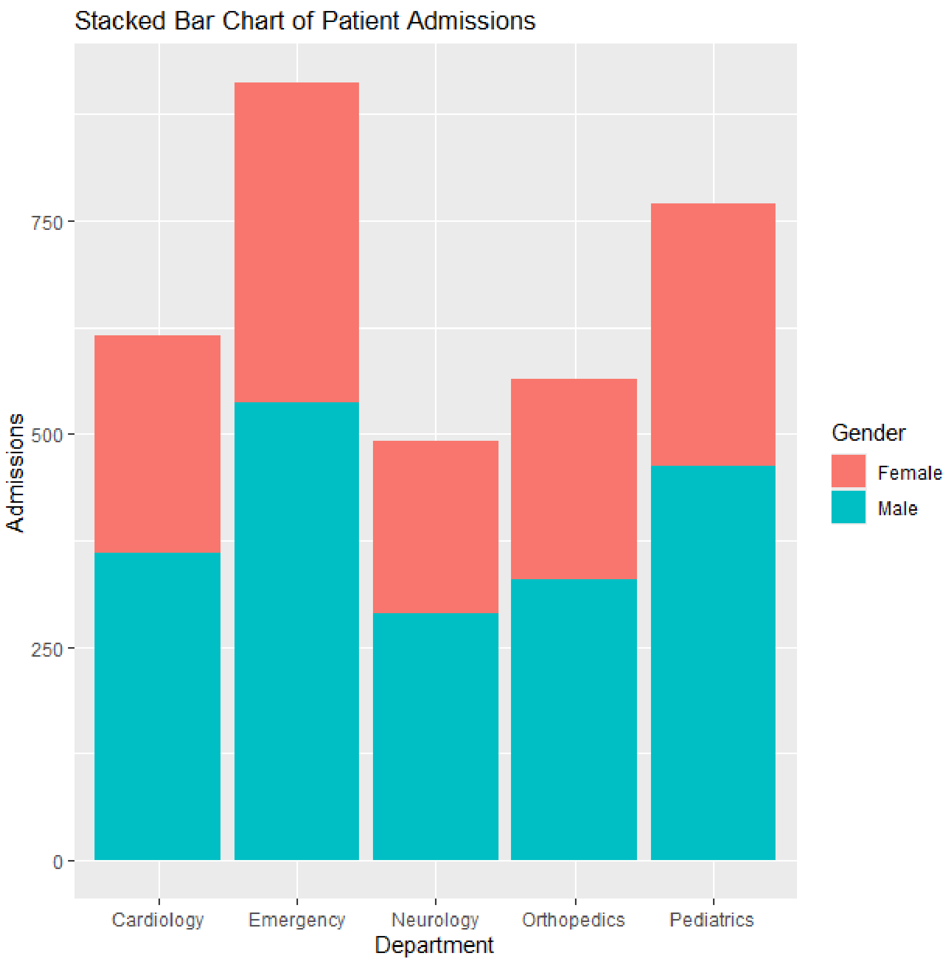
1. Bar Chart

```
ggplot(hospital, aes(x=Department, y=Admissions, fill=Department)) +
  stat_summary(fun=mean, geom="bar") +
  labs(title="Average Patient Admissions by Department",
       x="Department",
       y="Average Admissions")
```



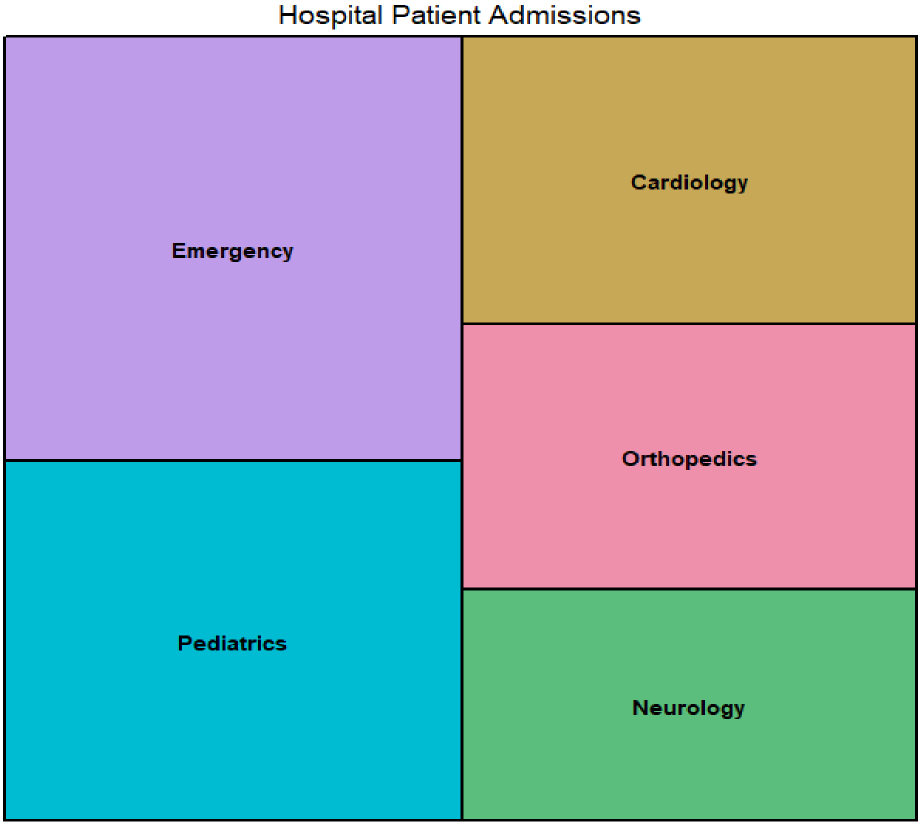
2. Stacked Bar Chart

```
ggplot(hospital, aes(x=Department, y=Admissions, fill=Gender)) +
  geom_bar(stat="identity") +
  labs(title="Stacked Bar Chart of Patient Admissions",
       x="Department",
       y="Admissions")
```



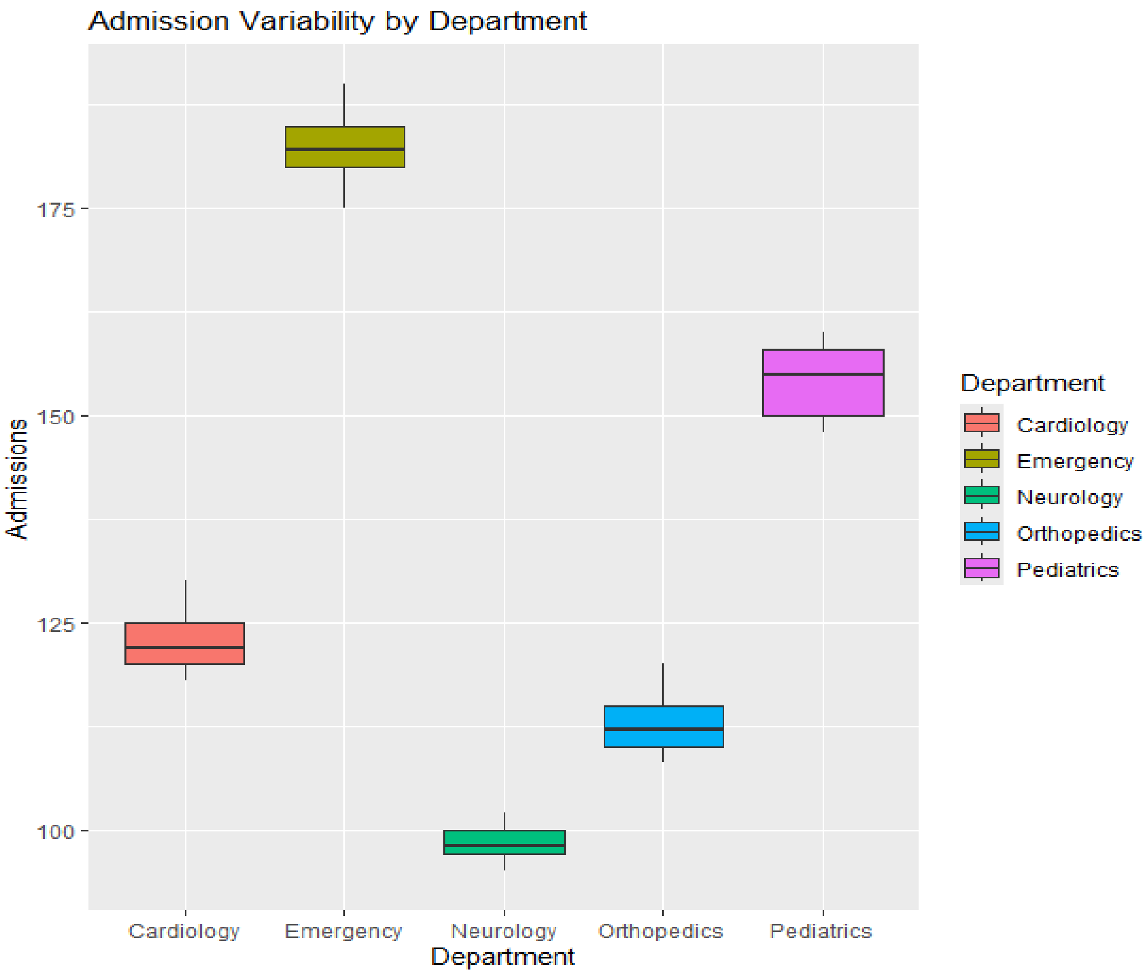
3. Treemap

```
dept <- aggregate(Admissions ~ Department, data=hospital, sum)
treemap(
  dept,
  index="Department",
  vSize="Admissions",
  title="Hospital Admissions Treemap"
)
```



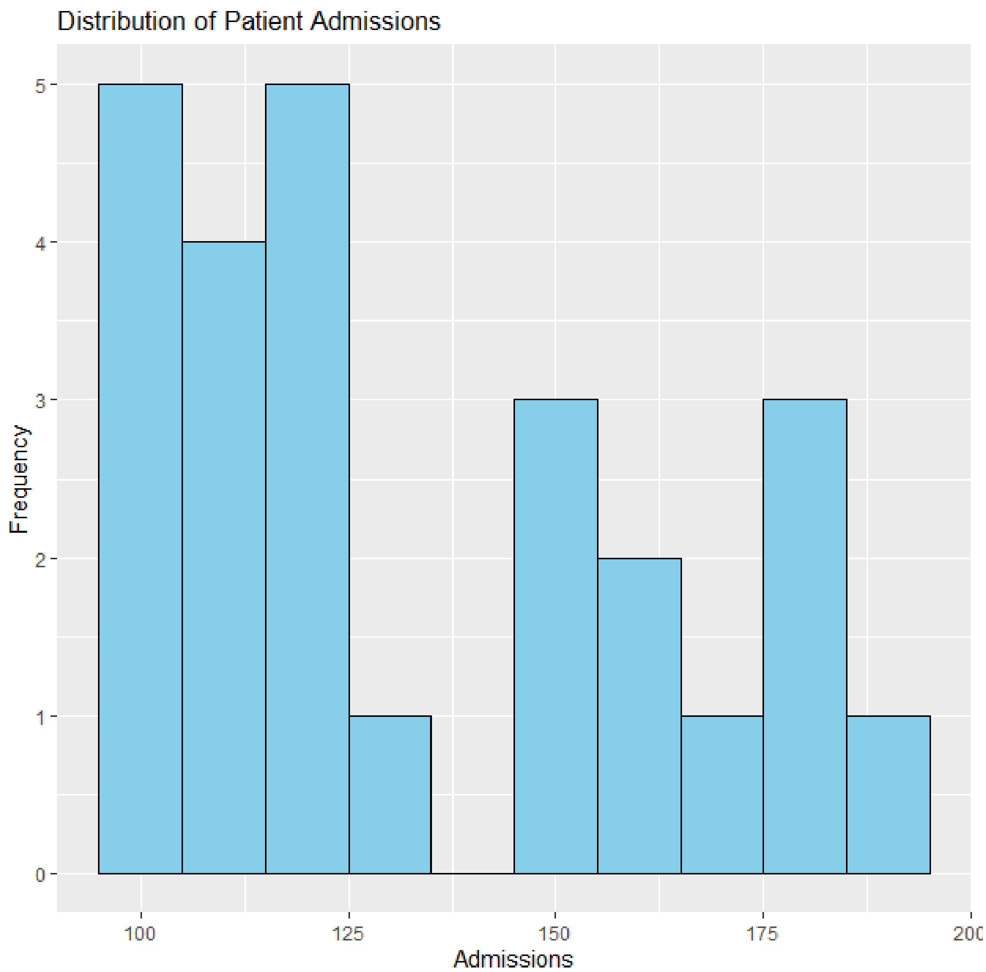
4. Box Plot

```
ggplot(hospital, aes(x=Department, y=Admissions, fill=Department)) +
  geom_boxplot() +
  labs(title="Admission Variability by Department",
       x="Department",
       y="Admissions")
```



5. Histogram

```
ggplot(hospital, aes(x=Admissions)) +
  geom_histogram(binwidth=10, fill="skyblue", color="black") +
  labs(title="Distribution of Patient Admissions",
       x="Admissions",
       y="Frequency")
```



SCENARIO 2: E-COMMERCE CUSTOMER SPENDING BEHAVIOUR

Dataset:

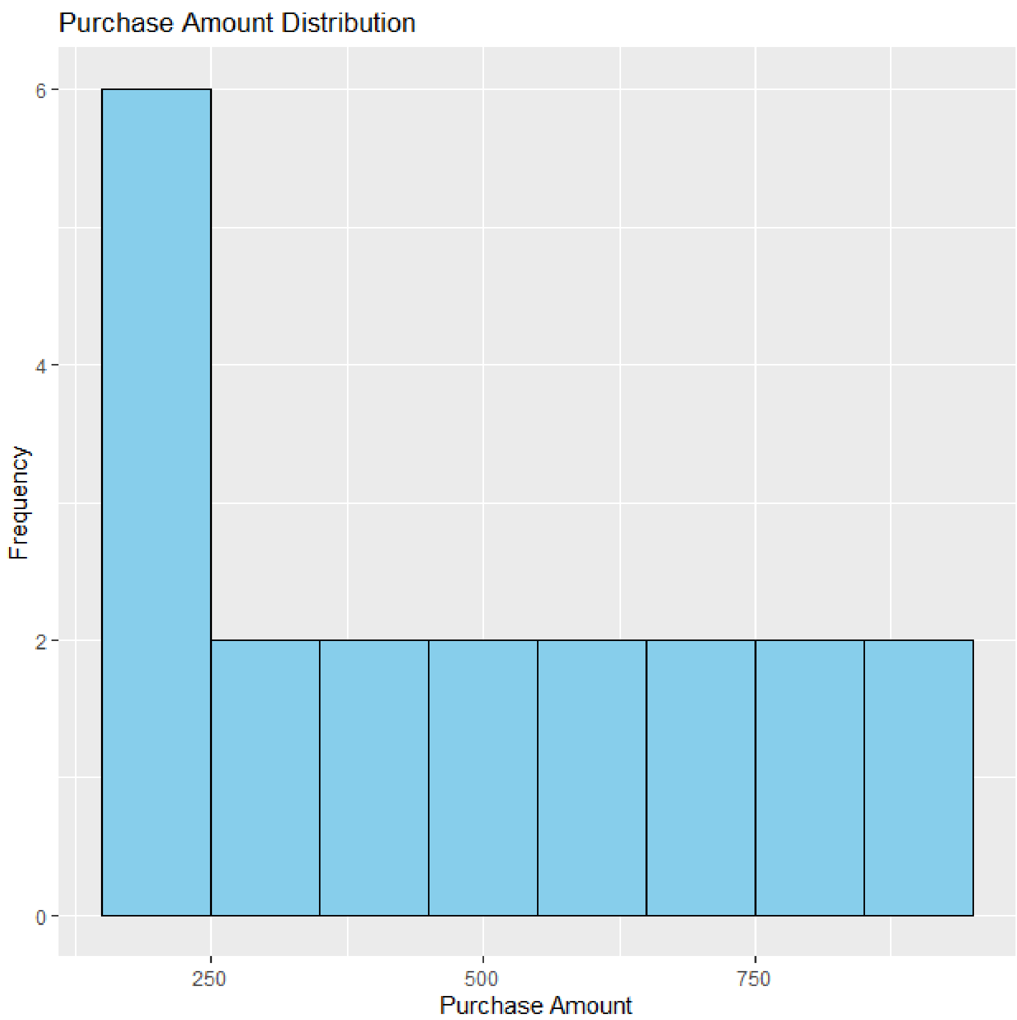
```
library(ggplot2)

customer <- data.frame(
 CustomerID = 1:20,
 Category = c(
  "Electronics","Clothing","Grocery","Furniture","Electronics",
  "Clothing","Grocery","Furniture","Electronics","Clothing",
  "Grocery","Furniture","Electronics","Clothing","Grocery",
  "Furniture","Electronics","Clothing","Grocery","Furniture"
 ),
 PurchaseAmount = c(
  500,200,150,900,700,
  350,180,750,650,400,
  220,850,550,300,170,
  800,600,450,210,950
 ),
 OrderFrequency = c(
```

```
    8,5,10,3,9,
    6,12,4,8,7,
    11,5,9,6,10,
    4,8,7,12,3
  )
)
```

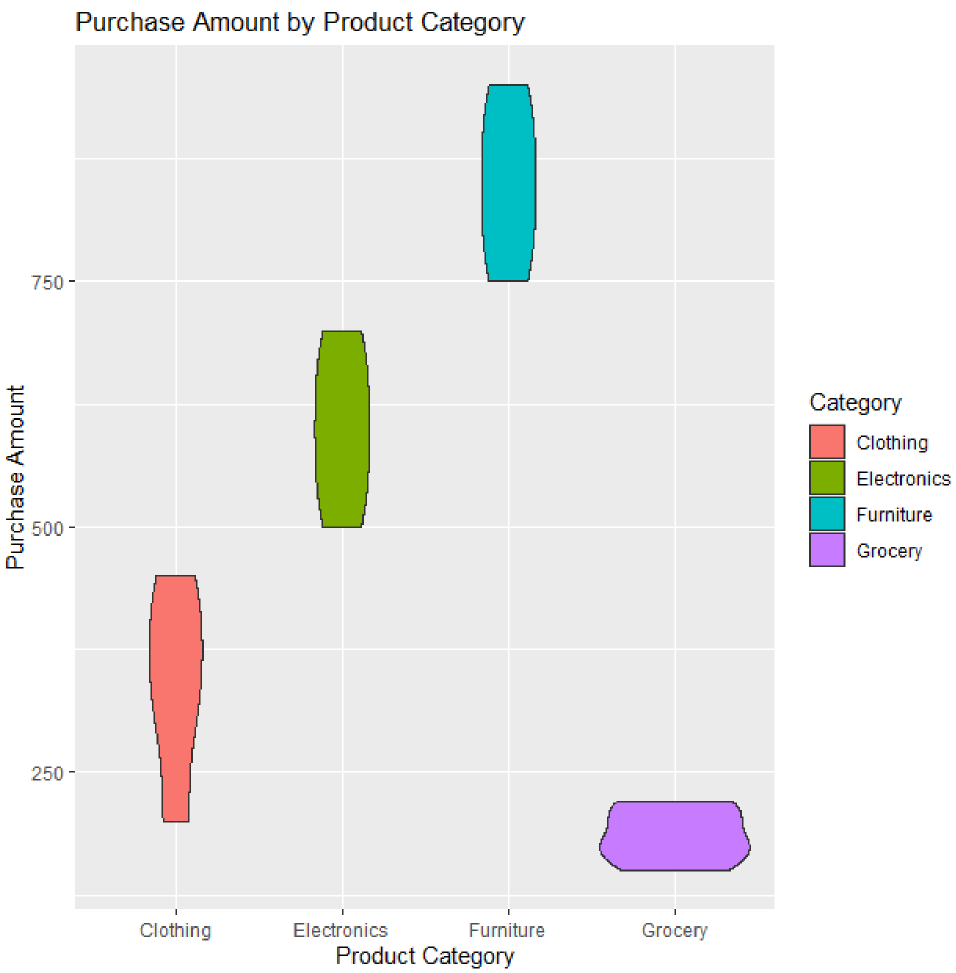
1. Histogram

```
ggplot(customer, aes(x = PurchaseAmount)) +
  geom_histogram(binwidth = 100, fill = "skyblue", color = "black") +
  labs(
    title = "Purchase Amount Distribution",
    x = "Purchase Amount",
    y = "Frequency"
  )
```



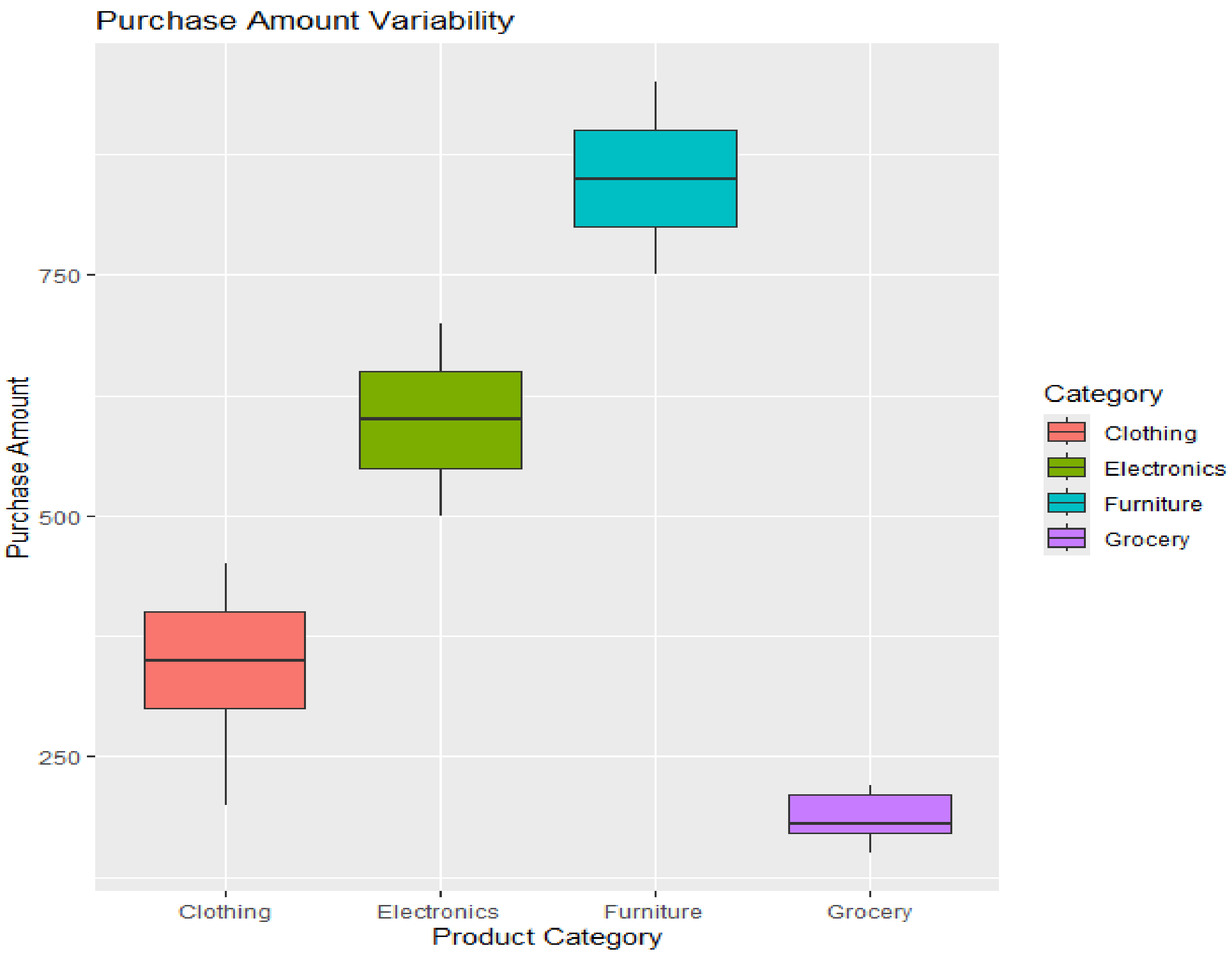
2. Violin Plot

```
ggplot(customer, aes(x = Category, y = PurchaseAmount, fill = Category)) +
  geom_violin() +
  labs(
    title = "Purchase Amount by Product Category",
    x = "Product Category",
    y = "Purchase Amount"
  )
```



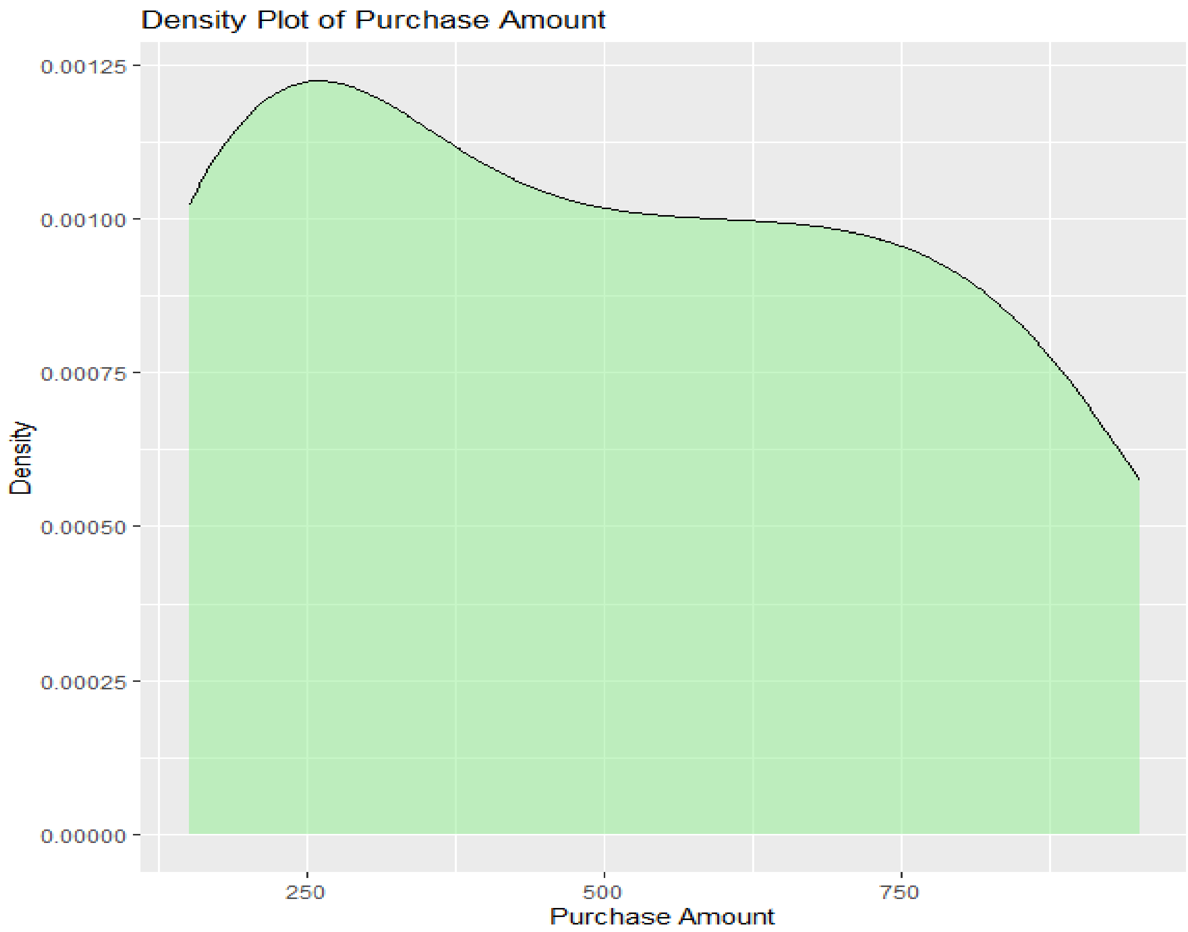
3. Box Plot

```
ggplot(customer, aes(x = Category, y = PurchaseAmount, fill = Category)) +
  geom_boxplot() +
  labs(
    title = "Purchase Amount Variability",
    x = "Product Category",
    y = "Purchase Amount"
  )
```



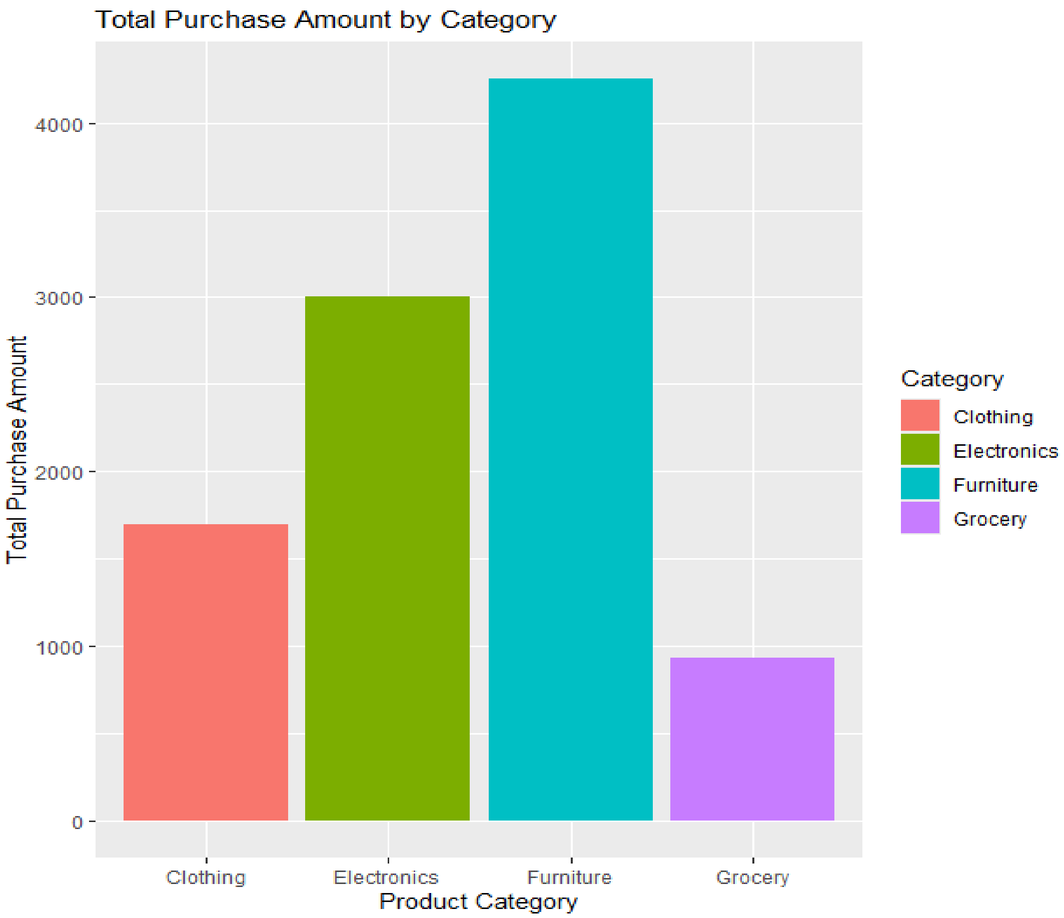
4. Density Plot

```
ggplot(customer, aes(x = PurchaseAmount)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  labs(
    title = "Density Plot of Purchase Amount",
    x = "Purchase Amount",
    y = "Density"
  )
```



5. Bar Chart

```
ggplot(customer, aes(x = Category, y = PurchaseAmount, fill = Category)) +
  stat_summary(fun = sum, geom = "bar") +
  labs(
    title = "Total Purchase Amount by Category",
    x = "Product Category",
    y = "Total Purchase Amount"
  )
```



SCENARIO 3: UNIVERSITY EXAMINATION PERFORMANCE DASHBOARD

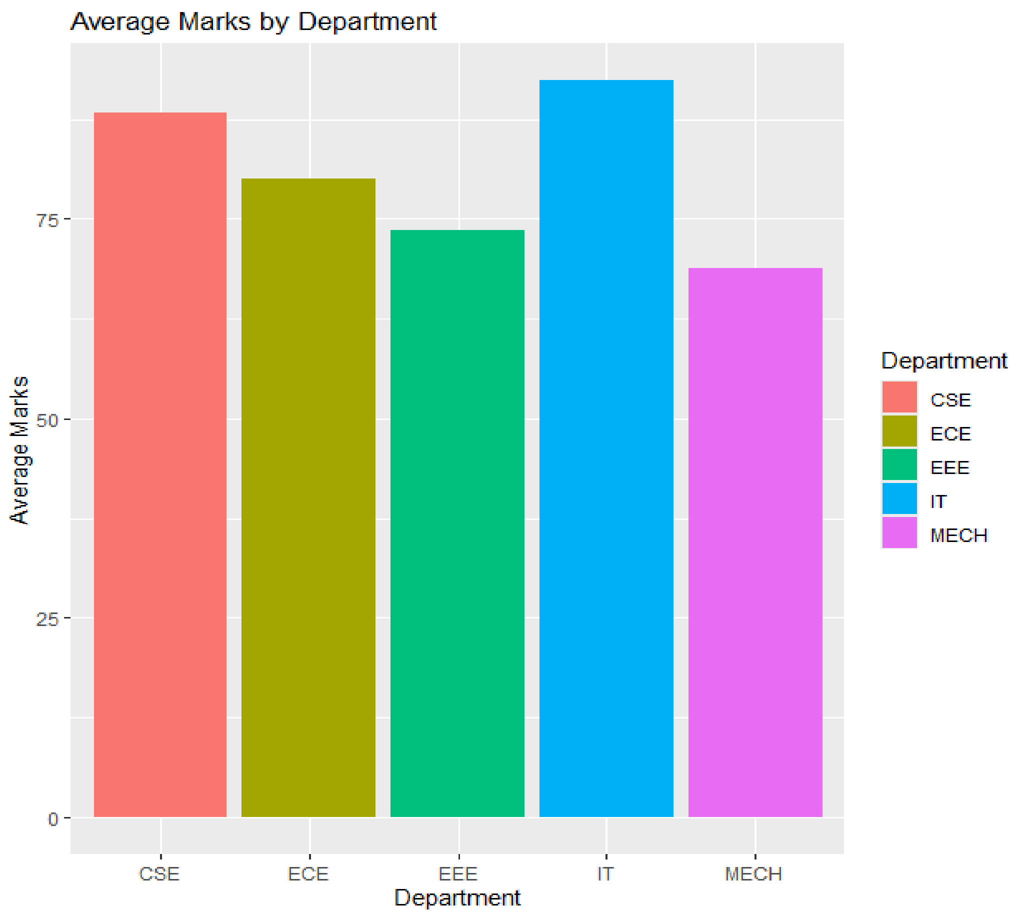
Dataset:

```
library(ggplot2)
student <- data.frame(
  Department = c(
    "CSE","CSE","CSE","CSE","CSE",
    "ECE","ECE","ECE","ECE","ECE",
    "EEE","EEE","EEE","EEE","EEE",
    "MECH","MECH","MECH","MECH","MECH",
    "IT","IT","IT","IT","IT"
  ),
  Class = c(
    "A","A","B","B","C",
    "A","A","B","B","C",
    "A","A","B","B","C",
    "A","A","B","B","C",
    "A","A","B","B","C"
  ),
  Marks = c(
```

```
    85,90,88,92,87,
    78,82,80,76,84,
    70,75,73,78,72,
    65,68,70,72,69,
    91,94,89,93,95
  )
)
```

1. Bar Chart

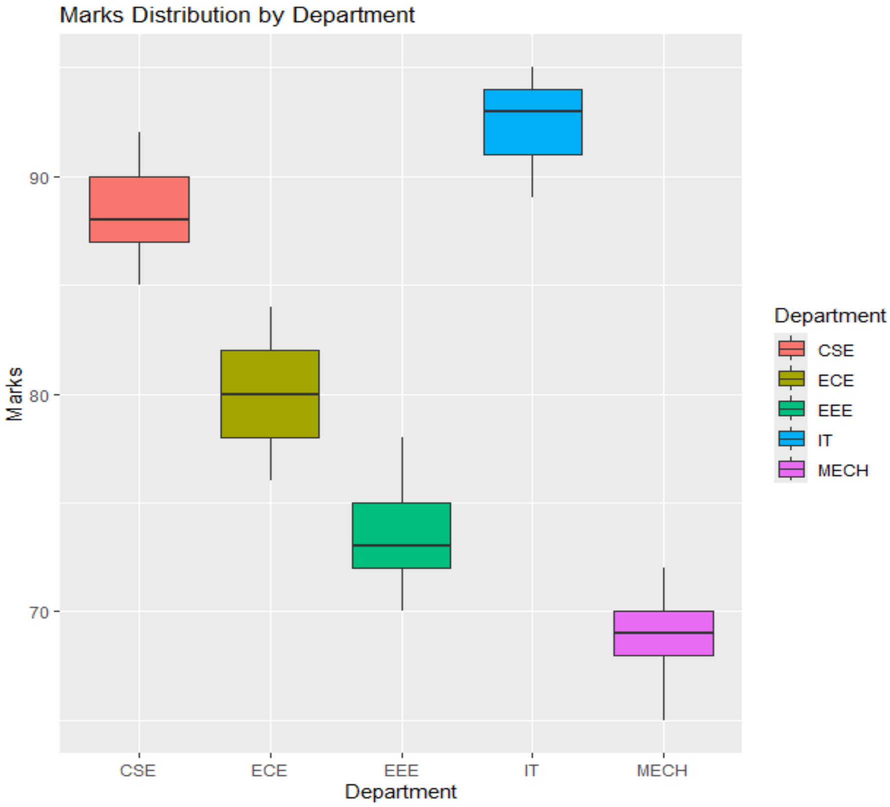
```
ggplot(student, aes(x = Department, y = Marks, fill = Department)) +
  stat_summary(fun = mean, geom = "bar") +
  labs(
    title = "Average Marks by Department",
    x = "Department",
    y = "Average Marks"
  )
```



2. Box Plot

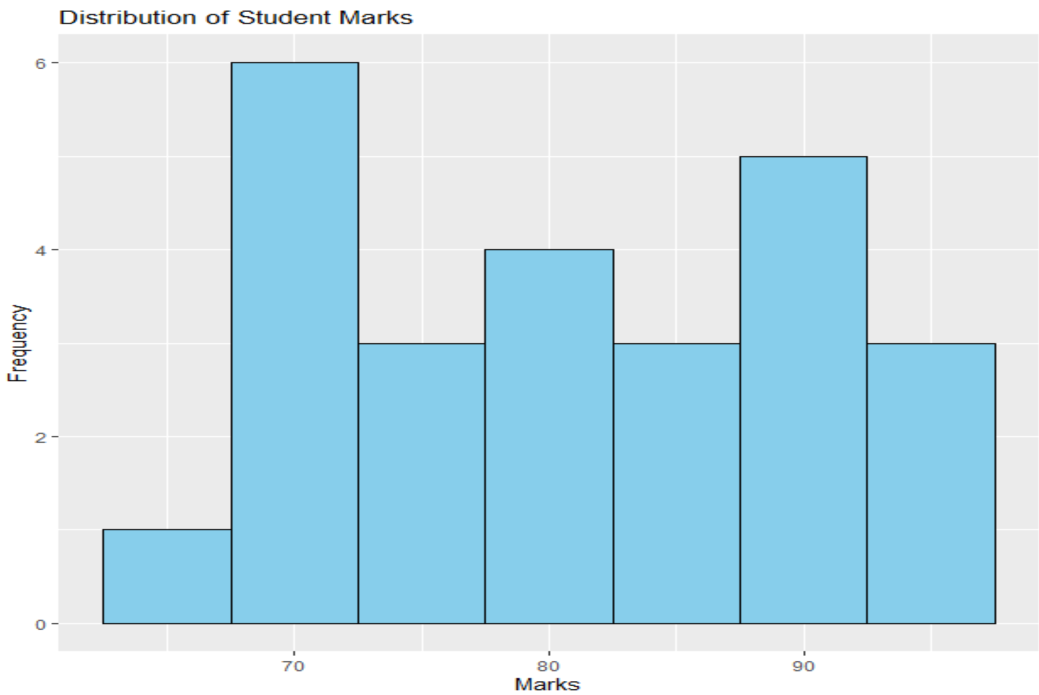
```
ggplot(student, aes(x = Department, y = Marks, fill = Department)) +
  geom_boxplot() +
  labs(
    title = "Marks Distribution by Department",
    x = "Department",
```

y = "Marks")



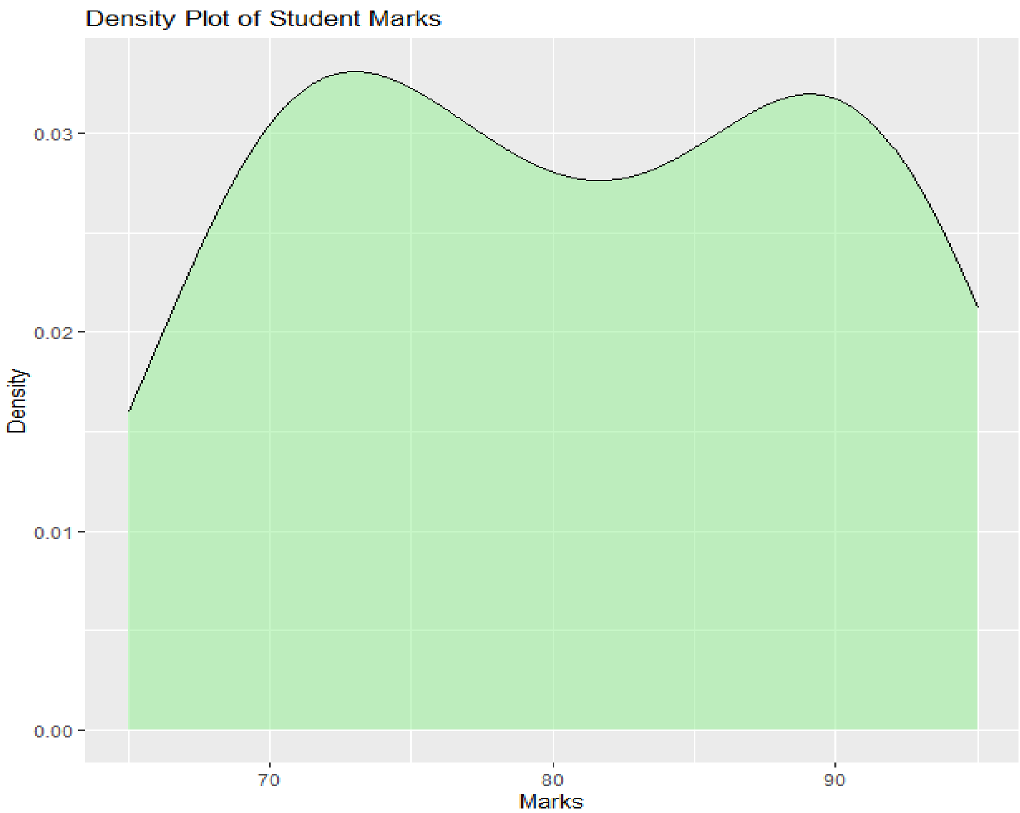
3. Histogram

```
ggplot(student, aes(x = Marks)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Student Marks",x = "Marks",y = "Frequency" )
```



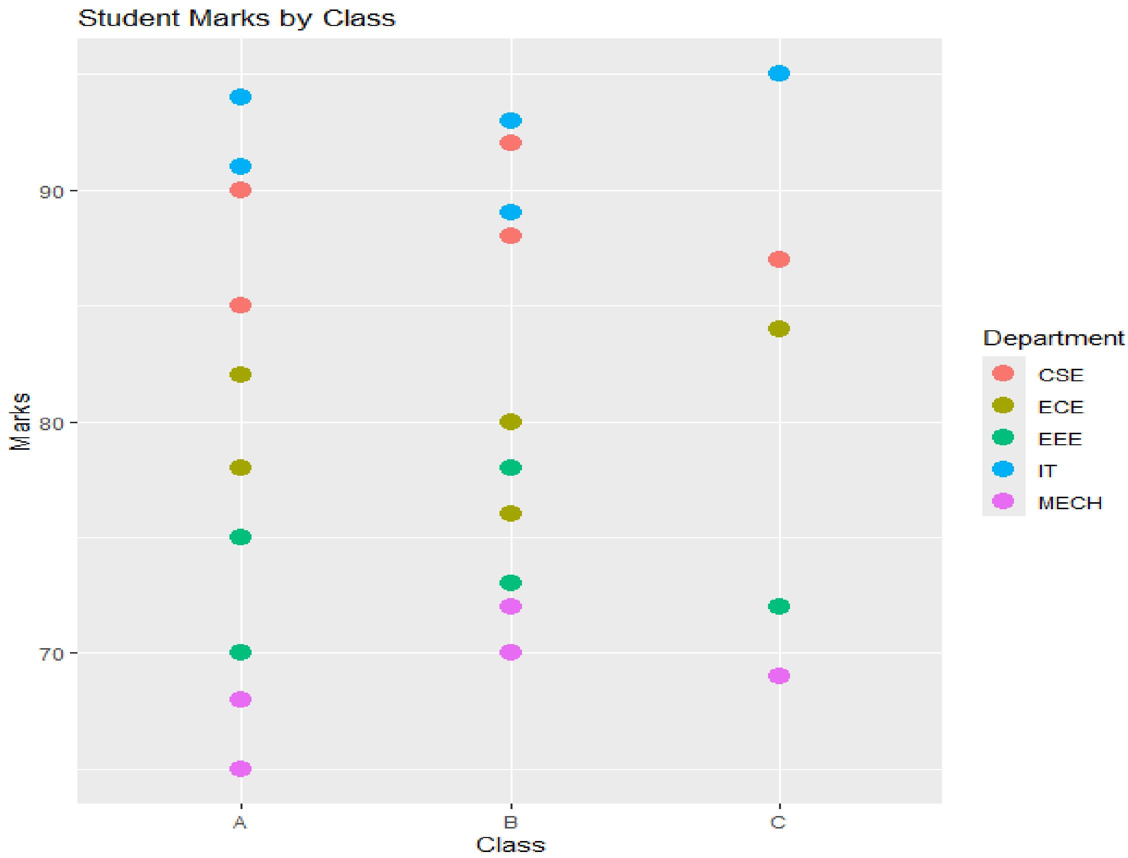
4. Density Plot

```
ggplot(student, aes(x = Marks)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  labs(
    title = "Density Plot of Student Marks",
    x = "Marks",
    y = "Density"
  )
```



5. Scatter Plot

```
ggplot(student, aes(x = Class, y = Marks, color = Department)) +
  geom_point(size = 4) +
  labs(
    title = "Student Marks by Class",
    x = "Class",
    y = "Marks"
  )
```



SCENARIO 4: CLIMATE AND RAINFALL DISTRIBUTION STUDY

Dataset:

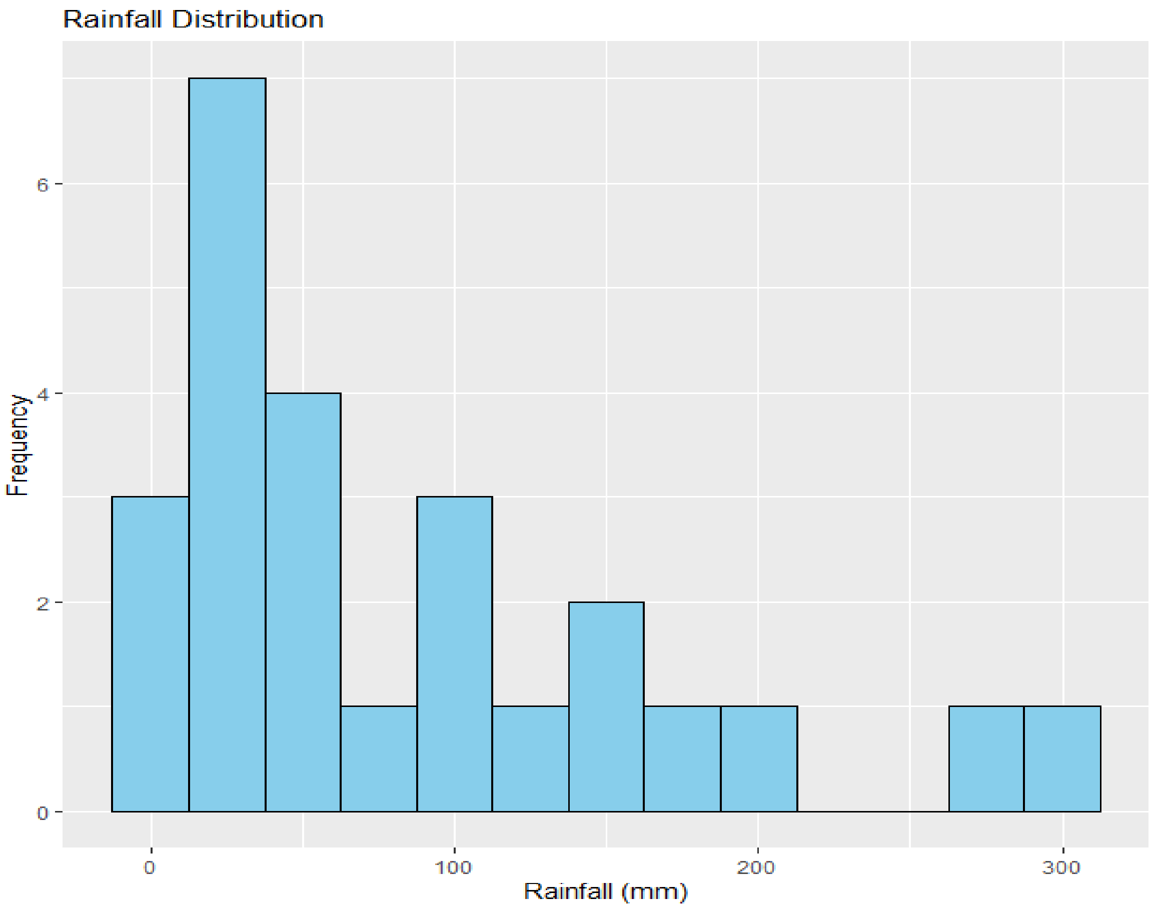
```
library(ggplot2)

climate <- data.frame(
 City = c(
  "Chennai","Chennai","Chennai","Chennai","Chennai",
  "Delhi","Delhi","Delhi","Delhi","Delhi",
  "Mumbai","Mumbai","Mumbai","Mumbai","Mumbai",
  "Bangalore","Bangalore","Bangalore","Bangalore","Bangalore",
  "Hyderabad","Hyderabad","Hyderabad","Hyderabad","Hyderabad"
 ),
 Month = c(
  "Jan","Mar","Jun","Sep","Dec",
  "Jan","Mar","Jun","Sep","Dec",
  "Jan","Mar","Jun","Sep","Dec",
  "Jan","Mar","Jun","Sep","Dec",
  "Jan","Mar","Jun","Sep","Dec"
 ),
 Rainfall = c(
  20,35,80,150,200,
```

```
    15,20,90,110,40,
    10,25,300,280,50,
    5,30,120,180,60,
    8,18,100,160,45
  ),
  Temperature = c(
    29,32,35,31,28,
    15,22,38,34,18,
    27,30,32,30,27,
    24,28,29,27,23,
    26,31,34,30,25
  ),
  Humidity = c(
    65,68,75,85,80,
    45,50,60,70,55,
    70,75,90,88,78,
    60,65,80,82,68,
    58,62,78,80,65
  )
)
```

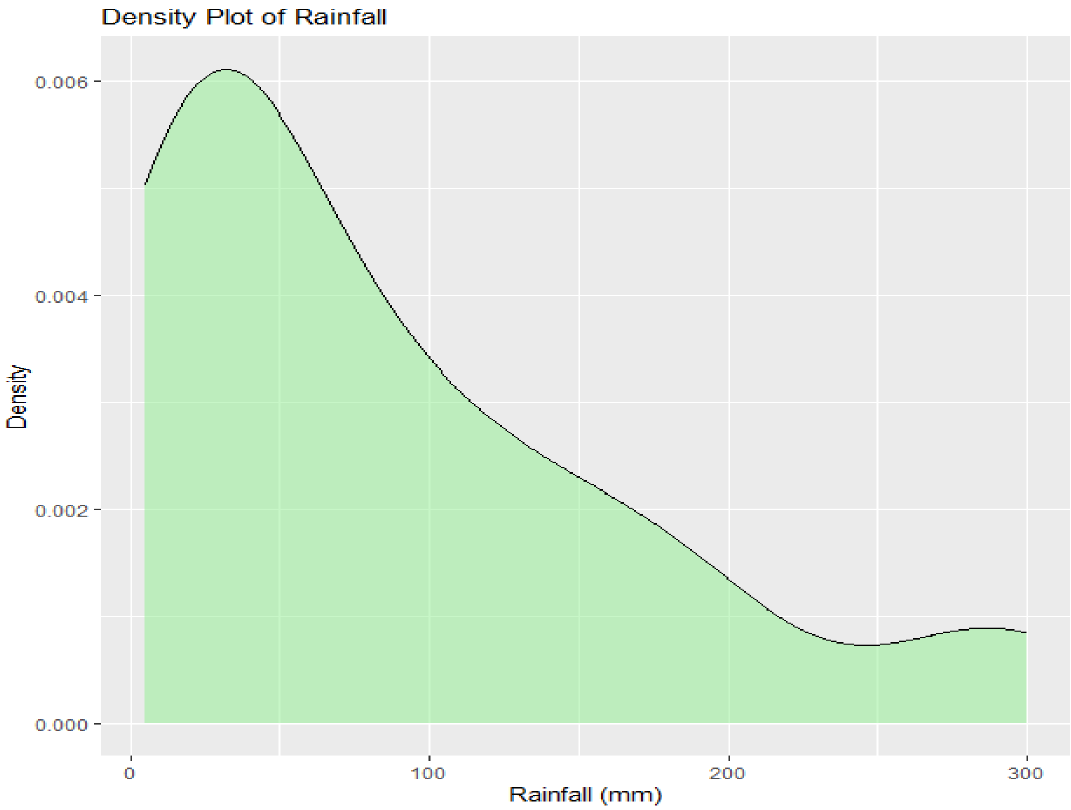
1. Histogram

```
ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 25, fill = "skyblue", color = "black") +
  labs(
    title = "Rainfall Distribution",
    x = "Rainfall (mm)",
    y = "Frequency"
  )
```



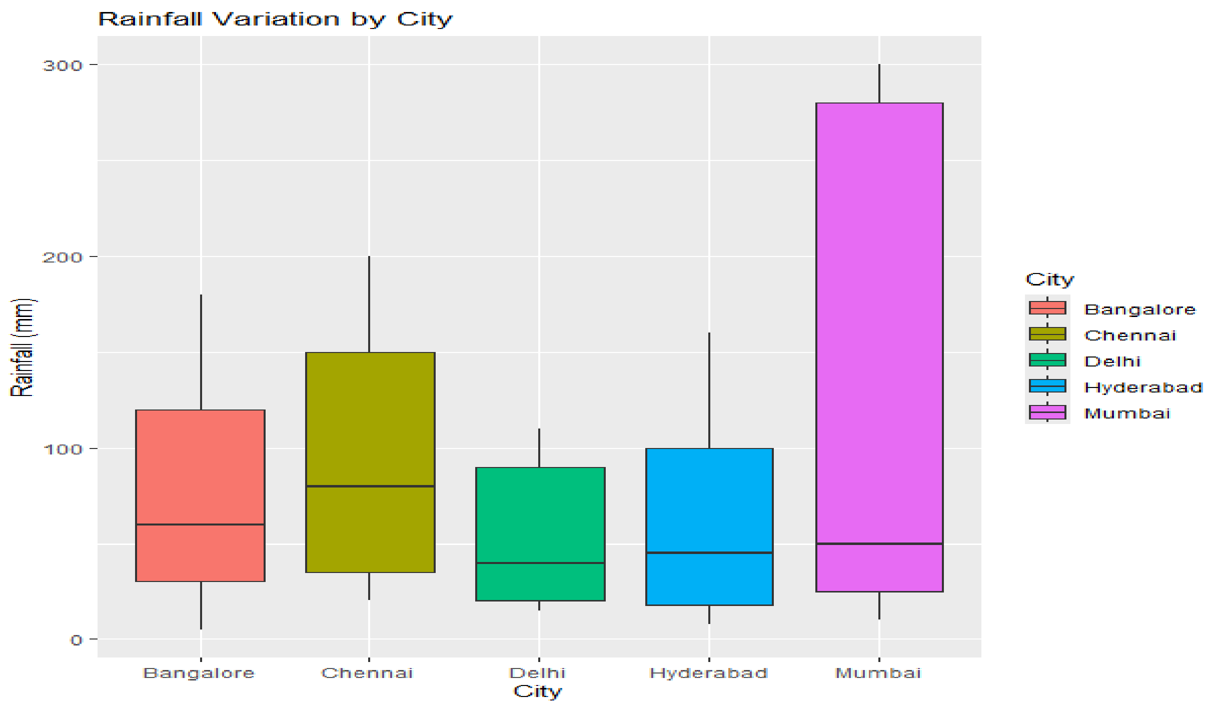
2. Density Plot

```
ggplot(climate, aes(x = Rainfall)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  labs(
    title = "Density Plot of Rainfall",
    x = "Rainfall (mm)",
    y = "Density"
  )
```



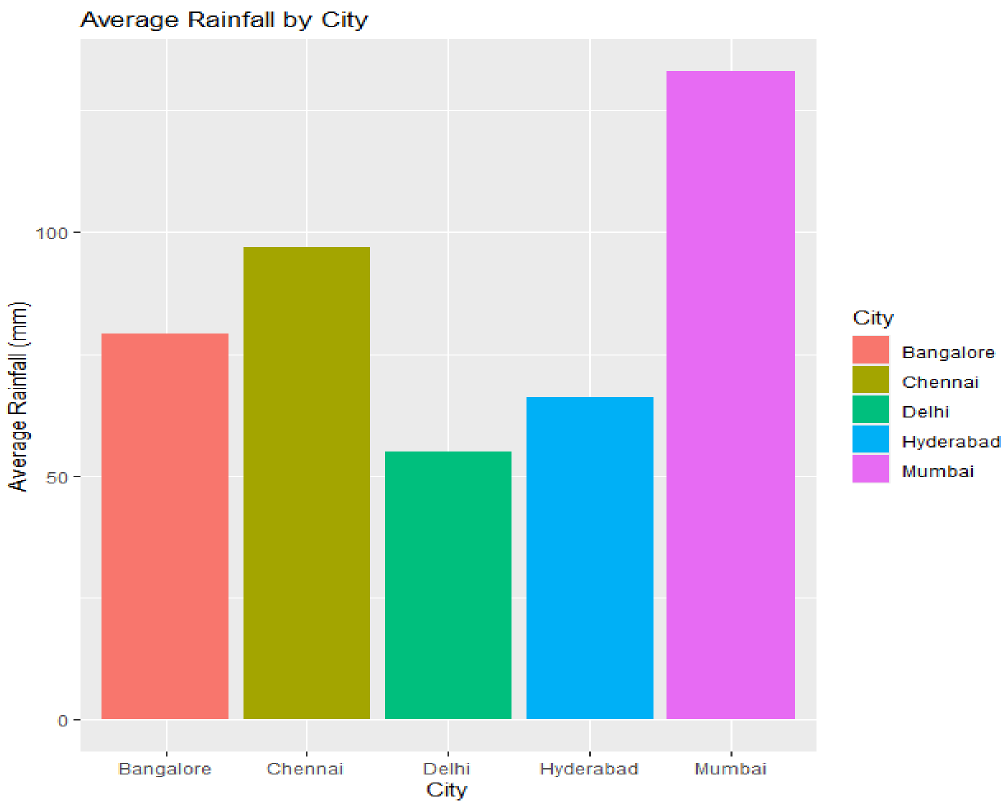
3. Box Plot

```
ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
  geom_boxplot() +
  labs(
    title = "Rainfall Variation by City",
    x = "City",
    y = "Rainfall (mm)"
  )
```



4. Bar Chart

```
ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +
 stat_summary(fun = mean, geom = "bar") +
 labs(
  title = "Average Rainfall by City",
  x = "City",
  y = "Average Rainfall (mm)" )
```



SCENARIO 5: BANK LOAN RISK ASSESSMENT

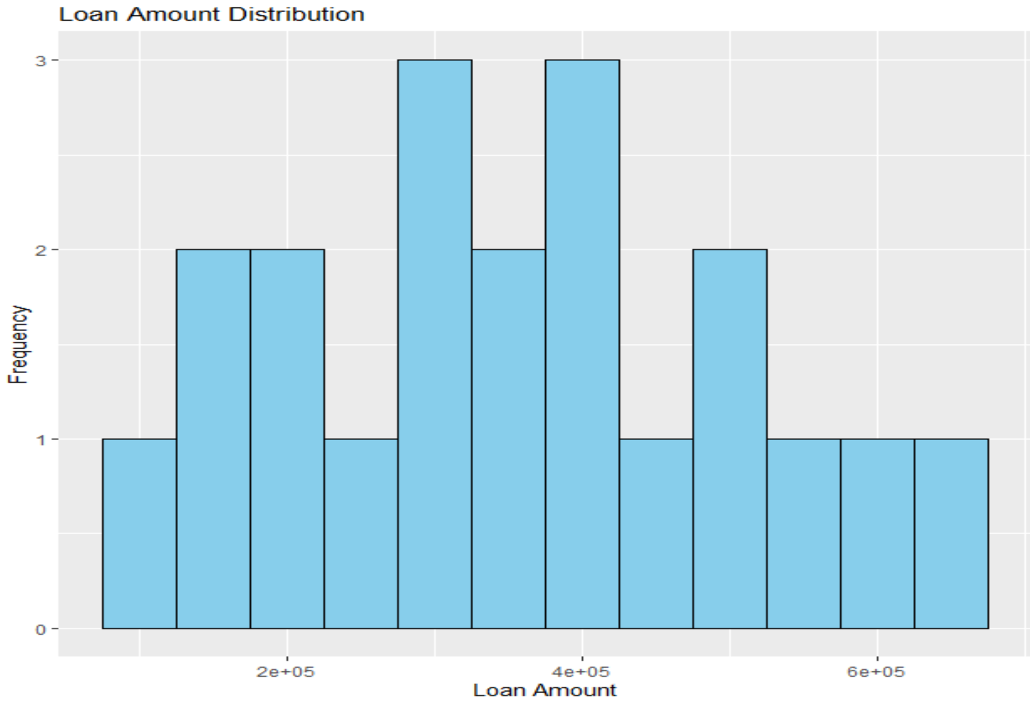
Dataset:

```
library(ggplot2)

loan <- data.frame(
  CustomerID = 1:20,
  Income = c(
    50000,60000,45000,80000,70000,
    35000,90000,55000,40000,65000,
    75000,30000,85000,48000,52000,
    68000,72000,38000,95000,58000
  ),
  CreditScore = c(
    750,780,680,820,790,
    620,850,710,650,760,
    800,600,830,690,720,
    770,810,640,860,730
  ),
  LoanAmount = c(
    300000,350000,200000,500000,420000,
    150000,600000,320000,180000,380000,
    450000,120000,550000,250000,280000,
    400000,480000,160000,650000,330000
  ),
  Repayment = c(
    "Good","Good","Average","Good","Good",
    "Poor","Good","Average","Poor","Good",
    "Good","Poor","Good","Average","Average",
    "Good","Good","Poor","Good","Average"
  )
)
```

1. Histogram

```
ggplot(loan, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 50000, fill = "skyblue", color = "black") +
  labs(
    title = "Loan Amount Distribution",
    x = "Loan Amount",
    y = "Frequency"
  )
```

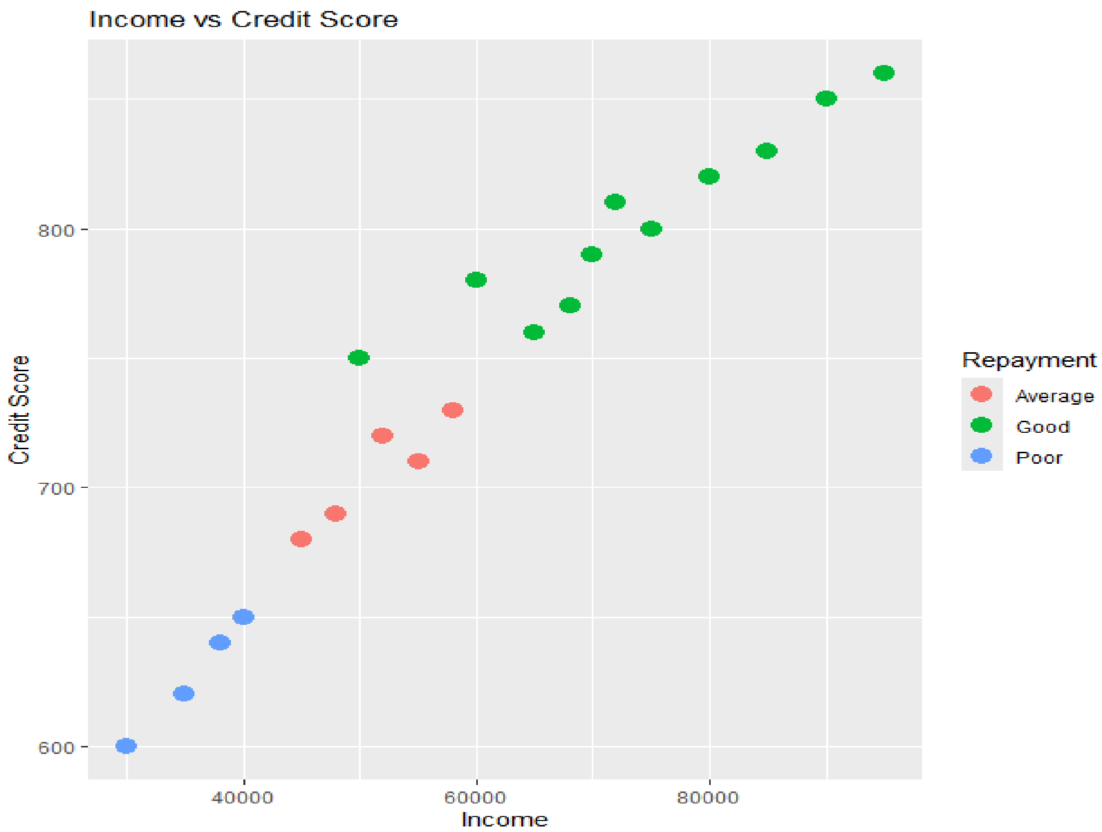


2. Box Plot

```
ggplot(loan, aes(x = Repayment, y = LoanAmount, fill = Repayment)) +
  geom_boxplot() +
  labs(
    title = "Loan Amount by Repayment History",
    x = "Repayment History",
    y = "Loan Amount"
  )
```

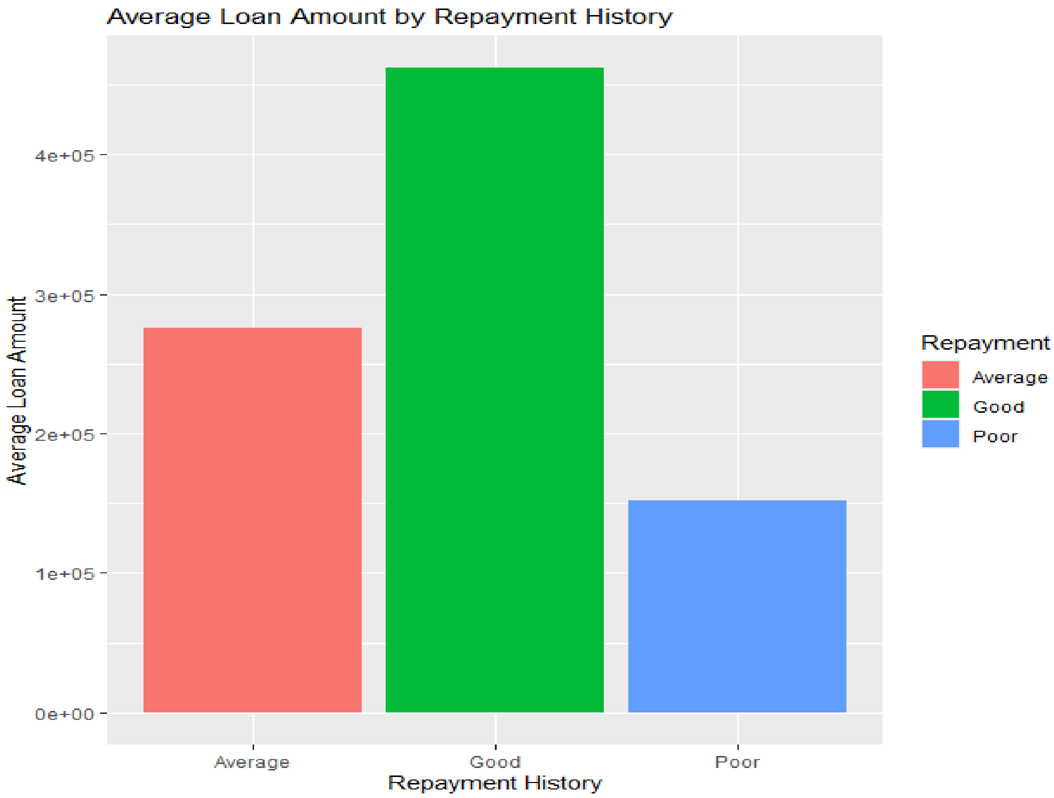
3. Scatter Plot

```
ggplot(loan, aes(x = Income, y = CreditScore, color = Repayment)) +
  geom_point(size = 4) +
  labs(
    title = "Income vs Credit Score",
    x = "Income",
    y = "Credit Score"
  )
```



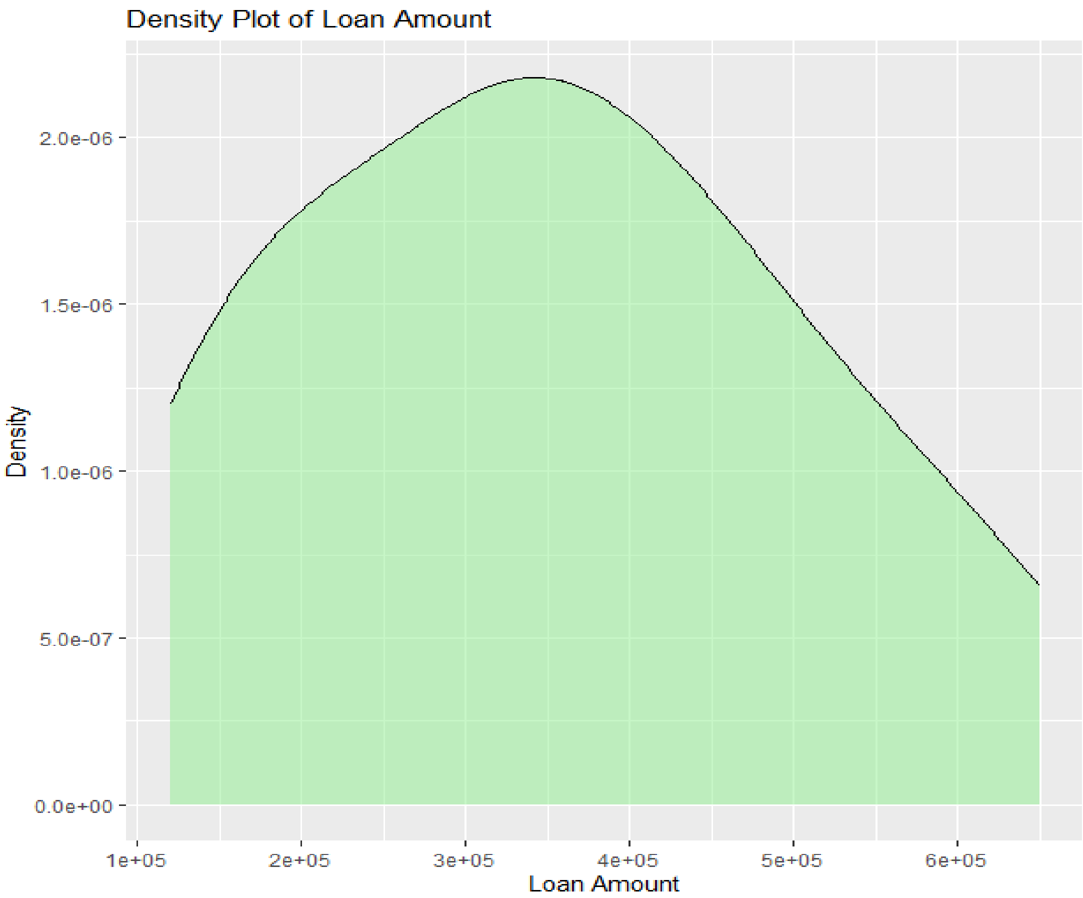
4. Bar Chart

```
ggplot(loan, aes(x = Repayment, y = LoanAmount, fill = Repayment)) +
  stat_summary(fun = mean, geom = "bar") +
  labs(
    title = "Average Loan Amount by Repayment History",
    x = "Repayment History",
    y = "Average Loan Amount"
  )
```



5. Density Plot

```
ggplot(loan, aes(x = LoanAmount)) +
 geom_density(fill = "lightgreen", alpha = 0.5) +
 labs(
  title = "Density Plot of Loan Amount",
  x = "Loan Amount",
  y = "Density"
 )
```


Density Plot of Loan Amount
Density
2.0e-06
1.5e-06
1.0e-06
5.0e-07
0.0e+00
1e+05
2e+05
3e+05
4e+05
5e+05
6e+05
Loan Amount